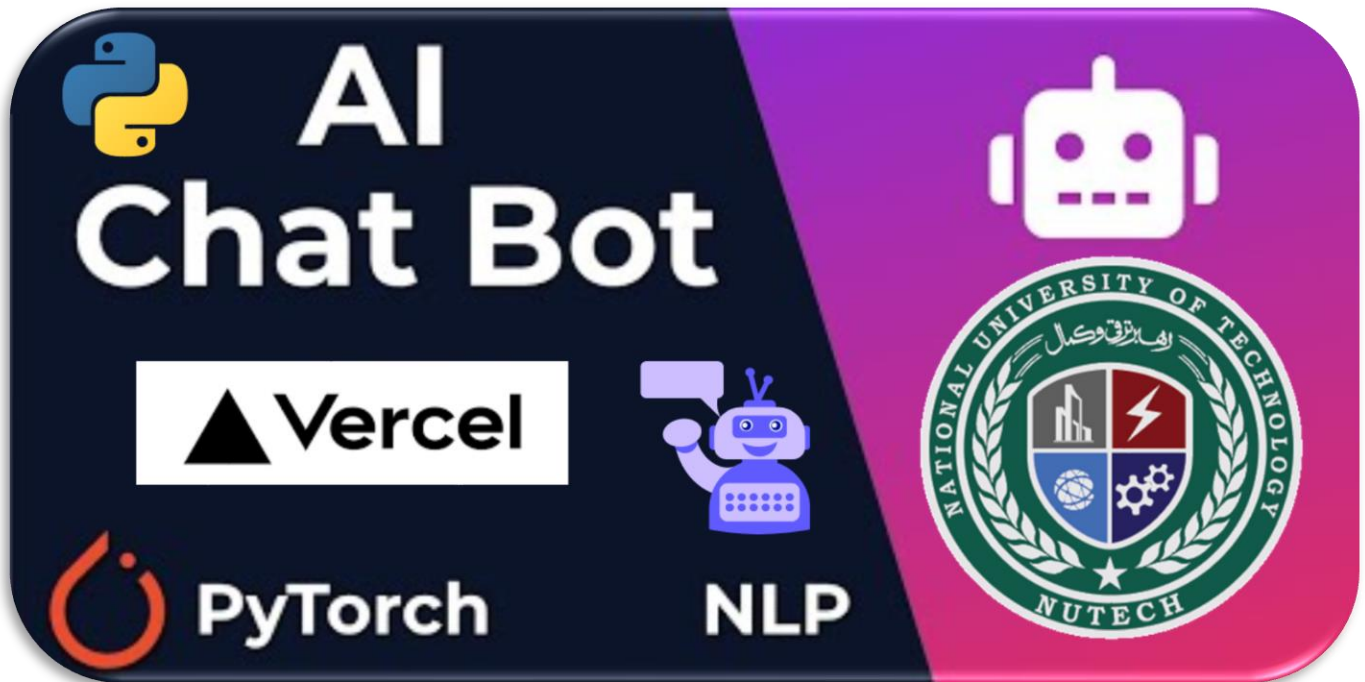


NUTECH ChatBot

Developed by NIC



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Introduction & Objectives

Introduction:

- The "Chatbot for NUTECH" project is an AI-driven solution designed to provide students, faculty, and prospective applicants with quick access to university-related information. This includes academic details, transport routes, admission inquiries, and event schedules etc, enhancing user experience and efficiency.

Objectives:

- Enhanced Information Accessibility:**
 - Ensure students have instant access to information without manual intervention.
- Automated Response System:**
 - Reduce the load on administrative staff by handling frequently asked questions automatically.
- Transport & Facility Information:**
 - Provide detailed information about university transport routes and facilities.
- User Engagement:**
 - Improve user interaction through a conversational interface tailored to the needs of NUTECH students.

Use Cases

Use Case	Description	Benefit
Admission Queries	The chatbot answers questions related to admission procedures, deadlines, and requirements.	Streamlines the admission process by providing instant information.
Campus Facilities	Provides details on university facilities such as libraries, labs, and cafeterias.	Helps students navigate and utilize campus resources efficiently.
Transport Information	Shares real-time updates on university transport routes, schedules, and any changes.	Ensures students can plan their commute effectively.
Event Information	Offers details about upcoming university events, seminars, and workshops.	Increases student participation by keeping them informed about events.

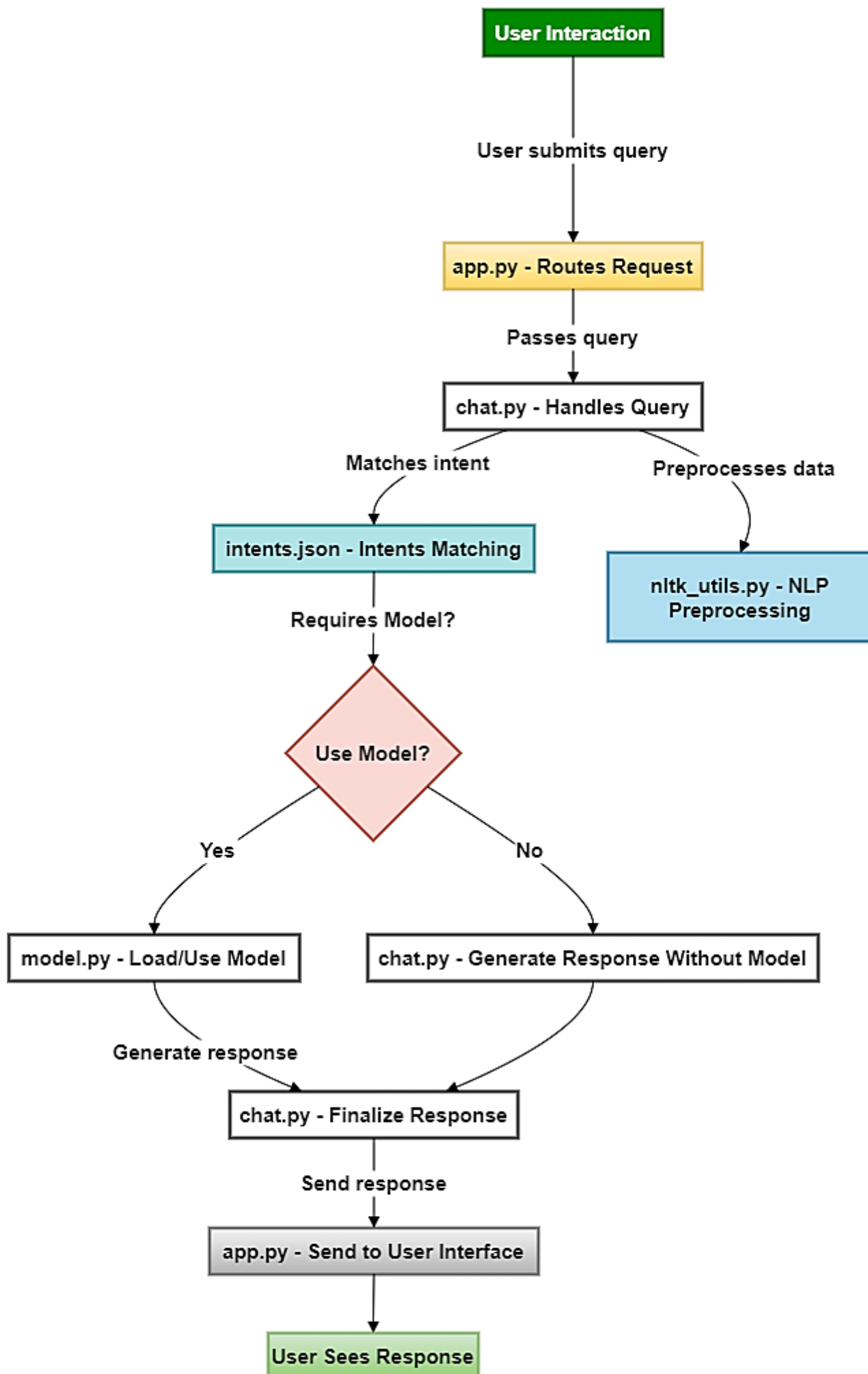
Project Structure & Workflow

Project Structure:

- Folder & Files:**

NUTECH-CHATBOT/: Main directory containing all project files.

- static/images/:** Stores static assets like images.
- templates/index.html:** HTML template for the web interface.
- app.py:** Entry point of the application, handling HTTP requests and responses.
- chat.py:** Contains logic for managing chatbot conversations.
- intents.json:** Defines intents and corresponding responses for the chatbot.
- model.py:** Handles loading and interacting with the language model.
- nltk_utils.py:** Utility functions for text processing.
- train.py:** Script for training the model.
- vercel.json:** Configuration for deployment on Vercel.



Workflow:

1. User Query:

- User interacts with the chatbot through a web interface.

2. Data Processing:

- Query is preprocessed using nltk_utils.py.
- The relevant intent is identified using intents.json.

3. Model Interaction:

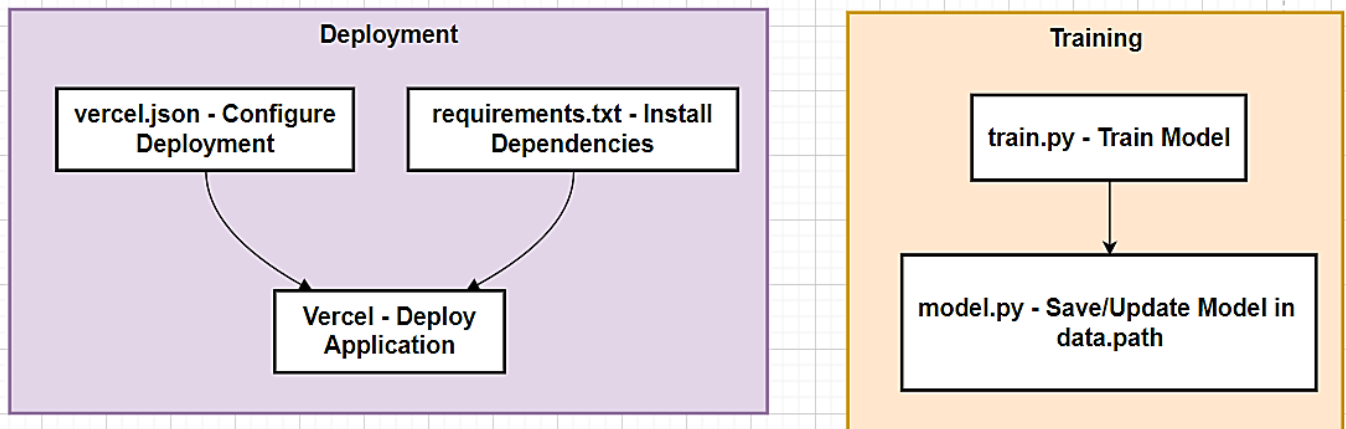
- The query is passed to the RAG model (model.py), which interacts with the phi3 and Llama-2 models to generate a response.

4. Response Generation:

- The generated response is fine-tuned and sent back to the user through chat.py.

5. Deployment:

- The application is deployed on Vercel using the configuration in vercel.json.



Model Working

Data Processing:

- **Web Scrapping:**
 - Data is scraped from the NUTECH website, including academic information, transport routes, and event details. Etc and stored in the JSON format

Model Interaction:

- **RAG Model:**
 - The Retrieval-Augmented Generation (RAG) model queries the vector database and generates a contextually relevant response.
 - Integrates with the phi3 language model for conversational accuracy.
- **Response Refinement:**
 - The response is further refined using Llama-2, ensuring it matches the tone and style expected by users.

Similarity Calculations:

- **Cosine Similarity:**
 - Scikit-learn's cosine similarity metrics are used to match the user query with the most relevant vectors, ensuring accurate information retrieval.
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Limitations

- **Accuracy Issues:**
 - The current model sometimes provides incorrect or irrelevant responses, especially for complex or nuanced queries.
 - **Resource Constraints:**
 - The current implementation cannot utilize more advanced models like Mistral 7B or 70B due to hardware limitations (no GPU support).
 - **Scalability:**
 - The use of JSON for intents might limit scalability as the number of possible user queries grows.
 - **Vector Embedding Accuracy:**
 - The embeddings used may not be as precise or contextually aware as newer models, affecting the overall performance.
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Proposed Improvements

- **Model Upgrade:**
 - Implement Mistral 7B or 70B models from Llama 3 and Hugging Face to enhance response accuracy and the quality of vector embeddings.
- **Hardware Upgrade:**
 - Invest in a system with GPU support to run more advanced models effectively, improving response time and accuracy.
- **Intent Handling:**
 - Transition from JSON-based intents to a more dynamic and scalable solution, possibly integrating a more robust natural language processing framework.
- **Performance Optimization:**
 - Fine-tune the model using more diverse training data and perform regular updates to the underlying algorithms to adapt to new types of queries.